

Every contract claims a source. Check it.

Bytematch · source verification for Portaldot V3

THE PROBLEM

0 of 242

contracts on the Portaldot V3 testnet had source anyone could check when the Hacker House began.

No Etherscan. No Sourcify. The official explorers index only the Substrate side.

A README is a claim.
A match is a check.

IN THE REPOSITORY

```
## Deployed contracts
| LendingMarket | 0x7e34…566b |
Source: contracts/LendingMarket.sol
```

Written once, on deploy day. Nothing keeps it true.

ON CHAIN, CHECKED BY BYTEMATCH

Not matched

The code at 0x7e34…566b is not what LendingMarket.sol compiles to today. The project's own SECURITY.md says it was deployed before later fixes.

Bytes in. A verdict out.

01

Read the chain

eth_getCode returns the deployed bytes. That is all the chain knows.

02

Rebuild

Compile the Solidity with the exact solc version read from those bytes.

03

Compare every byte

Exact: metadata too, so the source text is proven. **Partial**: same program, other comments or paths.

One repository in.
Every contract it deployed, checked.

`github.com/Junedkhan121/portalswap`

PortalSwapRouter · exact	PortalSwapFactory · exact	WPOT · exact
tPotStaking · exact	tPotFaucet · exact	4 × MockERC20 · exact

No settings, no contract names. Bytematch reads the repository without running any of its code, finds the Foundry and Hardhat projects, pins npm and submodule dependencies, and checks every address it mentions.

We ran it on the whole hackathon.

74
repositories, from just their URLs

47
of the 56 with contracts on V3 matched to source

102
matches: 49 exact, 53 partial

34
deployed contracts that don’t match their repo

Across the chain: 118 of 242 contracts can now be checked, up from zero.

Built for Portaldot V3, not ported to it.

What Portaldot does	What Bytematch does about it
Runs EVM bytecode through pallet-revive	Verifies the 237 EVM contracts with stock solc; detects the 5 PolkaVM ones
No contract-creation lookup, indexers start mid-chain	Finds each deploy transaction by searching historical state, about 3 s for 250k blocks
A new storage slot costs about 4.5 POT	SourceRoots: every result under one Merkle root, about 0.01 POT a contract
EXTCODEHASH is keccak of the EVM code	On-chain checks see the same hash the verifier computes, so contracts can gate on it

Every verification. One root on chain.

ONE SLOT PER CONTRACT

~9 POT

each. The 500 tPOT allowance covers about 54 contracts.

SOURCEROOTS

~0.01 POT

a contract. One Merkle root, overwritten on each update.

The entries travel in the event log, so anyone can rebuild every proof from the chain alone. Any contract calls check(target, attester, level, digest, proof) before it trusts a token.

Works where developers already are.

Foundry and Hardhat, unchanged

```
forge verify-contract … --verifier custom
--verifier-url …/api
```

Agents check before they trust

```
claude mcp add --transport http
bytematch …/mcp
```

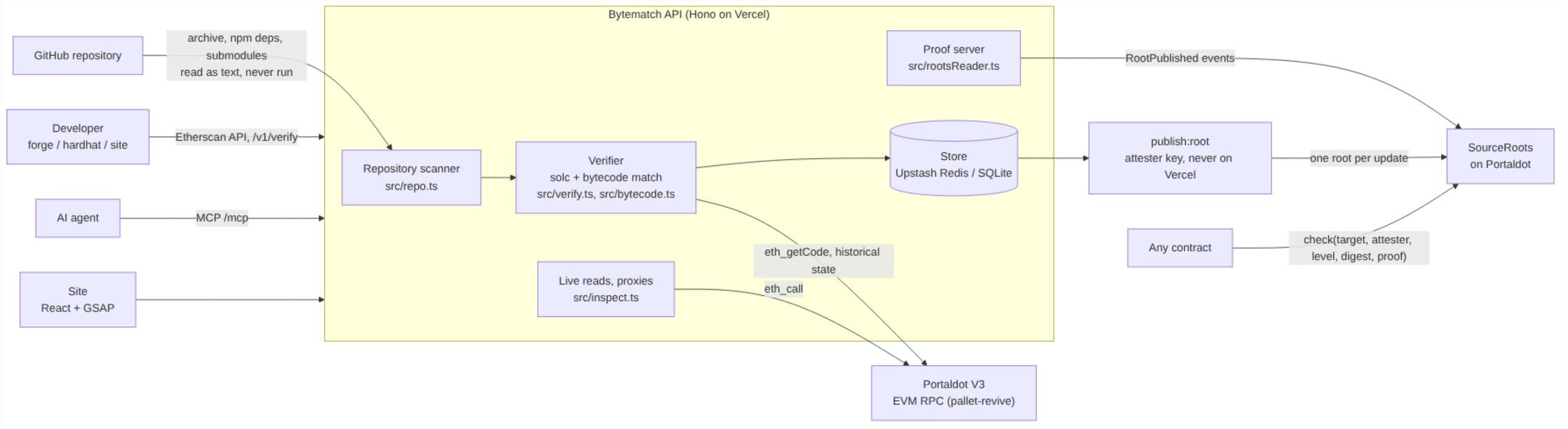
Read it live

Every public value, any view function, proxies read through their implementation.

Badge it

A README badge that shows the live match state, for every verified contract.

Architecture



On the Portaldot V3 testnet

Item	Address or transaction	Block
SourceRoots	`0x0f9691a6aae5d9b5ed7190551049ced18b8ed086`	306369
Current root, 100 programs, 0.033 POT	`0x8d2de9cf…cd6f9`	308025
Hello, deployed and verified live	`0xBDe6abB5a5075b65A522446FF2bF371148D53859`	308010
SourceRegistry (v1)	`0x6aea3b20e67e8b5b660d0d8eb917c790e190a203`	244720
17 attestations	`0xb0037034…3426 to 0xcab5cfe3…10b4`	244727–746
Attester	`0x8d375990766be8a0d1e1a93e6f9fa33ac2023ff3`	

Both contracts are verified by Bytematch itself as exact matches. Full hashes in the README.

Verification is what every other project needs first.

NEXT

- Mainnet, the day Portaldot V3 launches
- PolkaVM contracts compiled with resolc
- Verify every new deployment automatically
- Explorer and wallet integrations

WHO PAYS

- Public verification stays free
- Paid: private repositories and CI checks on every deploy
- Paid API for wallets, DEXs and explorers
- Audit firms comparing reports with deployed code

Check it.

bytematch-portaldot.vercel.app

github.com/miracle352/bytematch